



Gestation length and its associations with calf birth weight, calf perinatal mortality, and dystocia in dairy cattle

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ABSTRACT

The objective of the study was to quantify how nongenetic factors associate with gestation length; also of interest were the associations between gestation length and calving performance. The study comprised 9,750 gestation length records from 4,256 multiparous dairy cows, as few breeding dates were available for heifers. A linear mixed model was used to investigate factors associated with gestation length and logistic regression was used for the analysis of perinatal mortality (including stillbirths), calving assistance, and dystocia; calf birth weight and gestation length EBV were included as covariates in some models. The mean gestation length was 280.6 d (SD = 5.87), with twin births averaging 4.32 d (SE = 0.31 d) shorter than singleton births. Mean gestation length for dam parities 2, 3, 4, 5 to 7, and 8 to 12 were 278.43 d (SE = 0.64 d), 278.94 d (SE = 0.64 d), 279.40 d (SE = 0.64 d), 279.83 d (SE = 0.64 d), and 279.78 d (SE = 0.71 d), respectively. Of the 11 sire breeds represented in the data set, Holstein-Friesian had the shortest gestation length (277.3 d), and Limousin had the longest (282.82 d). The mean calf birth weight was 35.52 kg (SE = 0.44) for gestation lengths of 260 to 274 d, 37.09 kg (SE = 0.43) for 275 to 279 d, 38.27 kg (SE = 0.43) for 280 to 282 d, 39.37 kg (SE = 0.43) for 283 to 287 d, and 40.95 kg (SE = 0.44) for 288 to 305 d. Cows that conceived earlier (<49 DIM) had average gestation lengths 1.5 d longer than those that conceived after 132 DIM. Calves with very short (260 to 274 d) or very long gestations (288 to 305 d) had nearly double the odds of perinatal mortality relative to calves born with a gestation between 280 and 282 d. The need for calving assistance for calves born with longer gestation lengths (≥ 283 d) was 1.38 times that of calves born with gestation lengths of between 280 and 282 d. Additionally, the odds of dystocia for calves with very long

gestations (288 to 305 d) were almost 1.7 times those of calves born with gestation lengths of between 280 and 282 d. In conclusion, the study provides valuable insights for predicting expected calving dates by considering key factors associated with gestation length in cattle, such as sire breed, dam breed, genetic merit for direct gestation length, month of birth, DIM at conception, parity, twin status, and calf sex. Moreover, the study documents the associations between gestation length, calving difficulty, and perinatal mortality, while adjusting for calf birth weight and direct genetic merit for gestation length. Adjusting for these effects affected the risk of calving complications at extreme gestation lengths but did not explain all of the associations, especially for perinatal mortality. Understanding these factors can help explore strategies to optimize gestation length, such as selecting sires with desirable genetic merit for gestation length.

Key words: calf, calving, breed, twinning, reproduction

INTRODUCTION

Gestation length, defined as the period between conception and parturition, is a fundamental characteristic of mammals. Large interspecies variability exists in gestation length, ranging from just a few weeks in small rodents (Camín, 2010; Murray et al., 2010) to nearly 2 years in elephant species (Lueders et al., 2012; Pushpakumara et al., 2016). Differences in gestation length have also been reported among cattle breeds, with progeny sired by dairy breeds commonly averaging a gestation length of ~280 d and progeny from most beef-breed sires generally longer (Norman et al., 2009; Jourdain et al., 2024). Although gestation length in cattle typically spans a specific range of days, some variability does exist, which may be attributed to environmental factors. Several environmental factors have been documented to associate with gestation length in cattle. The factors include calf sex (Norman et al., 2009; Fitzgerald et al., 2015; Tomasek et al., 2017), dam parity (Norman et al., 2009; Fitzgerald et al., 2015; Tomasek et al., 2017), type of birth (singleton or multiple

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birth; Tomasek et al., 2017; Atashi and Asaadi, 2019), and month of calving (Tomasek et al., 2017; Vieira-Neto et al., 2017; Atashi and Asaadi, 2019). Understanding the contributions of these factors to gestation length variability can help producers make strategic decisions about drying off, grouping of animals on expected calving date, and better nutritional planning (Purfield et al., 2019).

Despite numerous studies having documented the association between nongenetic factors and gestation length in cattle (Norman et al., 2009; Tomasek et al., 2017; Vieira-Neto et al., 2017), none have investigated these relationships while simultaneously accounting for genetic merit for direct gestation length, a trait that is highly heritable. This could be particularly important if assortative mating is undertaken either directly on gestation length or indirectly via associated estimates of genetic merit such as fertility traits (Jenkins et al., 2016), calving traits (Eaglen et al., 2013), and milk yield (Norman et al., 2011). Moreover, no study has explored the association between gestation length and DIM at conception in cows. Given that dairy cows are often inseminated and conception established during lactation, it is important to deduce how this lactation phase at conception associates with gestation length. Additionally, although most previous studies have explored the associations between gestation length and all of perinatal mortality, calving assistance, and dystocia (Johanson et al., 2011; Atashi and Asaadi, 2019; Galluzzo et al., 2024), these analyses have generally not adjusted for differences in birth weight, which, in itself, is known to be associated with gestation length (Hansen et al., 2004; Chud et al., 2014; Jamrozik and Miller, 2014). Therefore, estimating the associations with gestation length independent of differences in birth weight may provide greater insights into the associations. The lack of available data on calf birth weight is likely the main limitation in conducting such adjusted analyses. Birth weight data are often not routinely collected or may be incomplete in Irish dairy commercial herds. However, research farms, with their well-maintained records and controlled environments, offer an opportunity to overcome this limitation. By leveraging data from research farms, the main objective of the present study was to quantify the factors associated with gestation length in dairy cattle. Also explored in this study were the associations between gestation length and calf birth weight, calving assistance, and perinatal calf mortality.

MATERIALS AND METHODS

Data

Phenotypic data used in the study were obtained from research trials on lactating dairy cows from 9

research farms between the years 1997 to 2022, inclusive. The available data set comprised 12,489 calving events from 5,001 cows bred with artificial insemination (AI) sires; gestation length records ranged from 260 to 305 d. Calvings recorded as abortions (births <260 d of gestation) were excluded from the study. Service dates and the sire of the calf were recorded for all lactations; however, due to limited service data for nulliparous heifers, gestations leading to the first calving were not included in this study. Data on calving dates, the degree of calving assistance, and calf birth weight were also available for all births. The degree of calving difficulty was recorded on a scale from 1 to 4: (1) no assistance, (2) some assistance, (3) considerable assistance, and (4) veterinary assistance. The recorded calving score was dichotomized for use in the present study into 2 separate categories: assistance (no assistance [1] vs. all other levels [2 to 4]) and dystocia (no assistance [1] or some assistance [2] vs. considerable [3] or veterinary assistance [4]). Additionally, perinatal mortality was recorded as a binary trait. In this study, perinatal mortality was defined as the death of the fetus or perinate before, during, or within 72 h of full-term calving (>260 d of gestation), including both stillbirths and early neonatal deaths (Mee et al., 2014). Birth weight was measured using a weighing scale. Only singleton and twin births were included; triplet or greater births were excluded.

Estimates of genetic merit for direct gestation length, represented as PTA values, from the national multibreed genetic evaluations in 2023 for all sires and maternal grandsires of the calves were available. Gestation length in Ireland is estimated using a multibreed, single-trait genetic evaluation, and all estimates of genetic merit (published as PTA) are expressed in days across breeds relative to a common base; an animal linear mixed model is used, with only a direct gestation length PTA estimated (<https://www.icbf.com/>). Only calves with a recorded sire and maternal grandsire were retained, and the calf genetic merit (i.e., EBV) for gestation length used in the present study was calculated as the sum of the PTA of the sire plus half the PTA of the maternal grandsire. Moreover, only parities where the AI sire recorded for the last insemination matched the sire recorded at the subsequent birth were considered.

After all edits, data were available on 9,750 calving events from 4,256 cows. The breed of sire was collapsed into Holstein-Friesian (6,372 calvings), Jersey (1,105 calvings), Scandinavian Red (526 calvings), Angus (958 calvings), Belgian Blue (197 calvings), Charolais (35 calvings), Hereford (80 calvings), Limousin (180 calvings), Montbéliarde (168 calvings), “other beef” (42 calvings), and “other dairy” (87 calvings). A total of 10 different dam breeds or breed-cross

categories were defined: Holstein-Friesian (6,799 calvings), Jersey (563 calvings), Holstein-Friesian × Jersey (889 calvings), Montbéliarde (163 calvings), Holstein-Friesian × Montbéliarde (218 calvings), Normande (74 calvings), Holstein-Friesian × Normande (70 calvings), Scandinavian Red (245 calvings), Holstein-Friesian × Scandinavian Red (430 calvings), and “other” (299 calvings). The heterosis coefficients of both the dam and the calf were calculated separately as per VanRaden and Sanders (2003), as

$$\text{Heterosis} = 1 - \sum_{i=1}^n \text{dam}_i \cdot \text{sire}_i, \quad [1]$$

where dam_i and sire_i represent the proportion of breed i in the dam and sire, respectively. Heterosis was categorized into 12 groups for both dam and calf. For dams, the groups were 0% (1,453 births), 1% to 10% (741 births), 11% to 20% (807 births), 21% to 30% (704 births), 31% to 40% (919 births), 41% to 50% (1,031 births), 51% to 60% (1,400 births), 61% to 70% (530 births), 71% to 80% (342 births), 81% to 90% (163 births), 91% to 99% (154 births), and 100% (1,509 births). For calves, the groups for heterosis were 0% (1,169 births), 1% to 10% (637 births), 11% to 20% (606 births), 21% to 30% (602 births), 31% to 40% (859 births), 41% to 50% (1,006 births), 51% to 60% (1,192 births), 61% to 70% (443 births), 71% to 80% (410 births), 81% to 90% (241 births), 91% to 99% (109 births), and 100% (2,479 births). Parity at calving was categorized as 2 (3,378 births), 3 (2,462 births), 4 (1,701 births), 5 to 7 (2,004 births), and 8 to 12 (208 births); DIM at conception were grouped as 22 to 49 d (1,117 births), 50 to 62 d (1,267 births), 63 to 76 d (2,135 births), 77 to 90 d (2,329 births), 91 to 104 d (1,539 births), 105 to 118 d (790 births), 119 to 132 d (412 births), and >132 d (434 births). Gestation length was categorized into 5 groups for later quantifying its relationship with birth weight, perinatal mortality, calving assistance, and dystocia; the categories were 260 to 274 d (1,301 births), 275 to 279 d (2,813 births), 280 to 282 d (2,146 births), 283 to 287 d (2,439 births), and 288 to 305 d (1,051 births). In addition, calf birth weight was classified into 5 categories: 15 to 30 kg (1,119 births), 31 to 35 kg (1,815 births), 36 to 40 kg (2,750 births), 41 to 45 kg (2,345 births), and >45 kg (1,721 births). Furthermore, calf gestation length EBV was classified into 5 groups as −8.96 to −5.50 d (1,747 calves), −5.49 to −3.52 d (3,077 calves), −3.51 to −2.20 d (1,907 calves), −2.19 to 0.00 d (2,192 calves), and >0.00 d (1,100 calves).

Statistical Analyses

A linear mixed model in the HPMIXED procedure in SAS version 9.4 (SAS Institute Inc., Cary, NC) was used

to explore the factors associated with differences in gestation length as follows:

$$\begin{aligned} Y_{cdijklmnpqr} = & \mu + DIM_i + Parity_j + Dam_k + Sire_l \\ & + HetC_c + HetD_d + Twin_m + Sex_n + Month_o + Year_r \\ & + Calf_p + CG_q + \varepsilon_{cdijklmnpqr}, \end{aligned} \quad [2]$$

where $Y_{cdijklmnpqr}$ = gestation length, μ = overall mean, DIM_i = effect of i th DIM class of the dam at conception, $Parity_j$ = effect of j th dam parity, Dam_k = effect of k th dam breed, $Sire_l$ = effect of l th sire breed, $HetC_c$ = effect of c th calf heterosis class, $HetD_d$ = effect of d th dam heterosis class, $Twin_m$ = effect of type of birth m (singleton or twin birth), Sex_n = effect of calf sex n (male or female), $Month_o$ = effect of o th month of calving (11 mo), $Year_r$ = effect of r th year of calving (1997 to 2022), $Calf_p$ = effect of calf p , CG_q = effect of q th contemporary group (herd-year-month of calving), and $\varepsilon_{cdijklmnpqr}$ = residual error term. Both cow and contemporary group were included as random effects, with the remaining factors treated as fixed effects. In a supplementary analysis, calf EBV for gestation length was included as a covariate in the model to quantify how phenotypic gestation length differed by EBV for gestation length.

A linear mixed model in the HPMIXED procedure in SAS was also used to explore whether gestation length, fitted as a fixed categorical effect, was associated with calf birth weight:

$$\begin{aligned} \text{Calf birth weight}_{cdijklmnpqrst} = & \mu + Parity_i + Dam_j \\ & + Sire_k + HetC_c + HetD_d + Twin_l + Sex_m + Month_n \\ & + Year_r + Gestation_s + DIM_t \\ & + Cow_o + CG_p + \varepsilon_{cdijklmnpqrst}, \end{aligned} \quad [3]$$

where, μ = overall mean, $Parity_i$ = effect of i th dam parity, Dam_j = effect of j th dam breed, $Sire_k$ = effect of k th sire breed, $HetC_c$ = effect of c th calf heterosis class, $HetD_d$ = effect of d th dam heterosis class, $Twin_l$ = effect of type of birth l (singleton or twin birth), Sex_m = effect of calf sex m (male or female), $Month_n$ = effect of n th month of birth (11 mo), $Year_r$ = effect of r th year of calving, $Gestation_s$ = effect of s th gestation length class, DIM_t = effect of t th DIM of the dam at conception, Cow_o = effect of the dam of the calf, CG_p = effect of p th contemporary group (herd-year-month of birth), and $\varepsilon_{cdijklmnpqrst}$ = residual error term. Both cow and contemporary group were included as random effects, with the remaining factors treated as fixed effects. The referent animal was a purebred Holstein-Friesian dam in its second parity that gave birth to a Holstein-Friesian-sired, female singleton calf in February 2022.

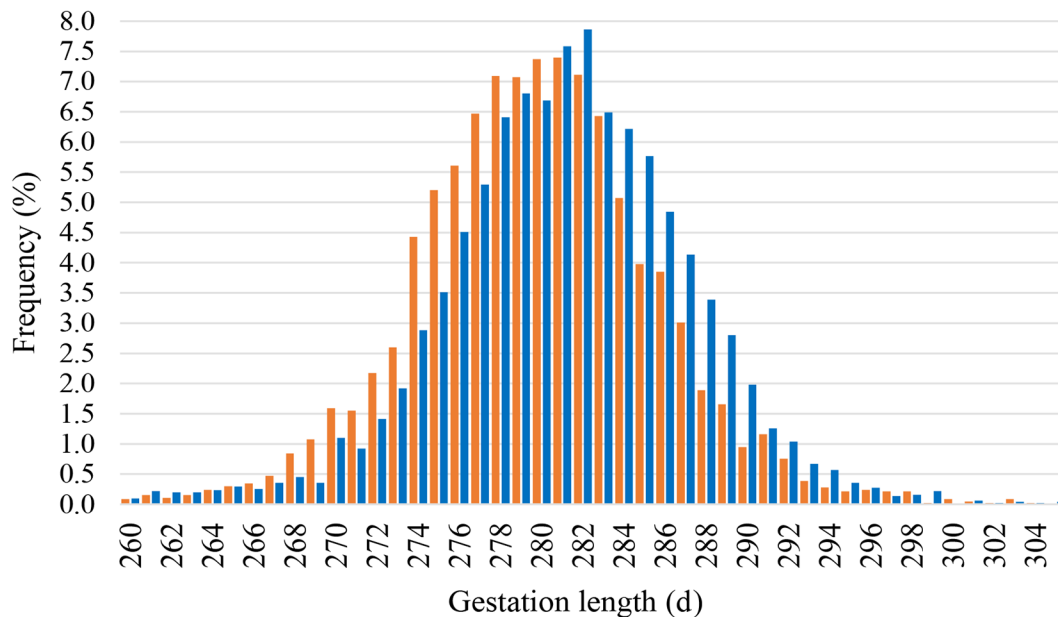


Figure 1. Distribution of gestation length for female (orange) and male (blue) calves.

When investigating whether gestation length was associated with the binary traits (i.e., perinatal mortality, calving assistance, or calving dystocia), logistic regression using the GENMOD procedure in SAS 9.4 with a logit link function while also accounting for the binomial distribution was used; the fixed effects in the model used were the same as those used for the analysis of birth weight. In an additional analysis, calf birth weight and gestation EBV were included as covariates in the models.

RESULTS

Summary Statistics

The mean gestation length of the entire data set was 280.6 d, with a raw standard deviation of 5.89 d. The mean incidence of calving assistance (some assistance, considerable assistance, and veterinary assistance), dystocia (considerable assistance and veterinary assistance), and perinatal mortality was 28.31%, 4.42%, and 4.03%, respectively. The incidence of twins was 2.61%. The mean birth weight of the calves was 39.09 kg with a SD of 7.08 kg; 52.3% of the calves born were males. The distribution of gestation length in the data set is shown in Figure 1; 92.5% of the gestation length records were between 270 and 290 d, and the median and mean DIM at the time of conception were 87 d and 90 d, respectively.

Factors Associated with Gestation Length

All factors considered in the mixed model analysis of gestation length were associated ($P < 0.05$) with gestation length. The predicted marginal means for the model effects of the sex of the calf, twin status, and parity are presented in Table 1. The gestation length of male calves was, on average, 1.10 d (SE = 0.09 d) longer than female calves. Twins were born, on average, 4.22 d (SE = 0.20 d) earlier than singletons. Calves from second-parity cows had the shortest predicted marginal mean for gestation length at 279.67 d (SE = 0.14 d). Mean gestation length gradually increased with parity to 280.30 d (SE = 0.16 d), 280.62 d (SE = 0.18 d), and 281.15 d (SE = 0.18 d) for parities 3, 4, and 5 to 7, respectively; gestation length did not change thereafter (Table 1). The association between dam parity and gestation length was consistent irrespective of whether the calf was a singleton or twin, or whether it was a male or female. Although both dam and calf heterosis were associated with gestation length, no obvious trend in their association by heterosis level was evident (Supplemental Table S1, see Notes). Cows that conceived earlier in their lactation cycle (between 22 and 48 DIM) had the longest predicted marginal mean for gestation length at 279.86 d (Figure 2); cows conceiving after 132 DIM had the shortest predicted marginal mean for gestation length at 278.25 d (Figure 2). The predicted marginal means for gestation length for

the calves of different sire and dam breeds are shown in Figures 3 and 4, respectively. Of the sire beef breeds, Angus had the shortest gestation length (278.6 d), and Limousin (284.7 d) had the longest (Figure 3). Also presented in Figure 3 is the breed mean PTA for gestation length based on the weighted average of the sires used in the study; there was good concordance between the phenotypic breed means and the mean gestation length PTA of the sires, although the mean phenotypic gestation length for dairy-sired calves was longer than expected based on their PTA. Of the dam dairy breeds, Normande, Normande cross, and Montbéliarde had the longest gestation length (Figure 4).

In a supplementary analysis including calf EBV for gestation length in the multiple regression model, the regression coefficient of gestation length on animal EBV for gestation length was 1.11 (SE = 0.03). The association between phenotypic gestation length and EBV for gestation length was consistent across calf sexes, dam parities, and sire breeds. Including gestation length EBV of the calf in the statistical model resulted in dam breed no longer being associated with phenotypic gestation length; the strength of the association between sire breed and gestation length reduced, with the *F*-statistic reducing from 33.83 when EBV was not in the model to 9.37 when gestation length EBV was in the model.

Birth Weight

Female calves were, on average, 36.86 (SE = 0.42) kg at birth, which was, on average, 2.8 kg lighter than their male counterparts (39.62 kg; SE = 0.42 kg). Twins (35.11 kg; SE = 0.45 kg) were individually lighter than singletons (41.37 kg; SE = 0.41). Calves from second-parity cows had the lightest predicted marginal mean for birth weight at 37.46 kg (SE = 0.47 kg), which gradually increased with parity; the heaviest birth weight was recorded for cows in parities 5 to 7, at 39.30 kg (SE = 0.48 kg; Supplemental Table S3, see Notes). Predicted marginal means for birth weight, as determined from the multiple regression model, increased with gestation length (Supplemental Table S3). In addition, the predicted marginal means for birth weight exhibited a subtle yet consistent upward trend as the number of days between calving and conception increased (Figure 2). When EBV for gestation length was then also included in the model for the association between birth weight and gestation length, every day longer in EBV for gestation length was associated with 0.28-kg (SE = 0.04) heavier birth weight. Including gestation length EBV in the multiple regression model reduced the average birth weight difference between the shortest and longest phenotypic gestation length categories from 5.33 kg to 4.88 kg (Supplemental Table S3).

Table 1. Predicted marginal means (SE in parentheses) for gestation length (d) for dam parity, calf sex, and twin status of the calf with or without adjustment for gestation EBV in the model

Effect and level	N	Predicted marginal means (SE)	
		Without adjustment for gestation EBV	With adjustment for gestation EBV
Parity			
2	3,378	278.43 ^c (0.64)	277.11 ^c (0.60)
3	2,462	278.94 ^b (0.64)	277.52 ^b (0.61)
4	1,701	279.40 ^b (0.64)	277.99 ^b (0.61)
5 to 7	2,004	279.83 ^a (0.64)	278.29 ^a (0.61)
8 to 12	208	279.78 ^{ab} (0.71)	278.01 ^{ab} (0.68)
Sex			
Female	4,652	278.73 ^b (0.64)	277.22 ^b (0.61)
Male	5,101	279.83 ^a (0.64)	278.34 ^a (0.61)
Twin status			
Twin	449	277.17 ^b (0.66)	275.61 ^b (0.63)
Singleton	9,304	281.38 ^a (0.63)	279.96 ^a (0.63)

^{a-c}Mean values within the same model term with different superscripts differ ($P < 0.05$).

Associations with Perinatal Mortality, Calving Assistance, and Dystocia

Odds ratios from the logistic regression analyses exploring the association between gestation length and perinatal mortality, calving assistance, and dystocia are shown in Figure 5. Relative to calves born with a gestation length of 280 to 282 d, the odds of perinatal mortality were almost double for those born with very short (260 to 274 d; odds ratio = 1.94) or very long (288 to 305 d; odds ratio = 1.73) gestations (Figure 5). No great change occurred in the odds ratios after adjusting for both calf birth weight and gestation length EBV, as the odds of perinatal mortality remained almost double for gestations of 260 to 274 d and gestations of 288 to 305 d relative to the reference category (280 to 282 d; Supplemental Table S2, see Notes). Nonetheless, both gestation EBV and calf birth weight were associated ($P < 0.05$) with the odds of perinatal mortality. A 1-unit increase in gestation EBV was associated with a 0.14 (SE = 0.04) increase in the logit of perinatal mortality, whereas a 1-kg increase in calf birth weight was associated with a 0.04 (SE = 0.01) reduction in the logit of perinatal mortality when both variables were included simultaneously in the model (Supplemental Table S4, see Notes).

The odds of requiring calving assistance was on average 1.3 times that of calves born following long gestations (283 to 287 d; odds ratio = 1.16 and 288 to 305

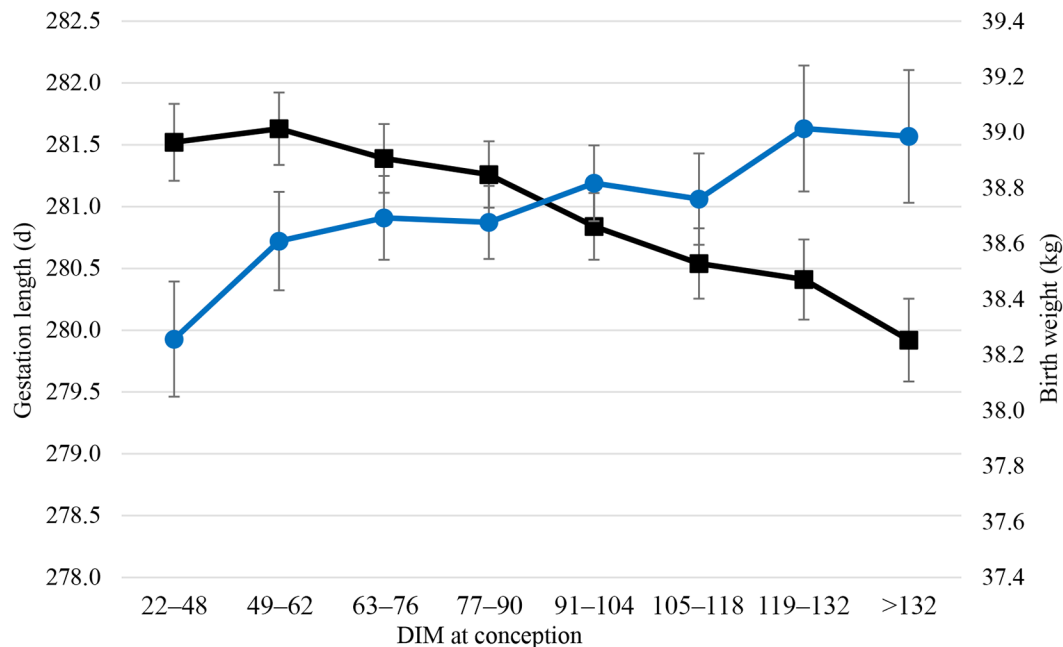


Figure 2. Trends in predicted marginal means for gestation length (black squares) and birth weight (blue circles) across varying DIM at conception (error bar extending 1 SE on either side of the mean).

d; odds ratio = 1.38) relative to those with gestations of between 280 and 282 d. A reduction in odds for requiring calving assistance for long gestations occurred once birth weight was adjusted for in the model (Supplemental Table S2). Moreover, the logit of calving assistance increased by 0.08 (SE = 0.01) and 0.07 (SE = 0.02) per 1-kg increase in calf birth weight and day of gestation EBV, respectively (Supplemental Table S4).

The odds of experiencing calving dystocia were almost double for those with very long gestation length (288 to 305 d; odds ratio = 1.7) relative to calves born with a gestation length of 280 to 282 d. The logit for dystocia increased by 0.09 (SE = 0.01) and 0.11 (SE = 0.04) per 1-kg increase in calf birth weight and day of gestation EBV, respectively, when individually included in the model. Although gestation length EBV was associated with dystocia, its inclusion did not lower the risk of dystocia for gestations between 288 and 305 d. However, adjusting for birth weight did reduce the risk of dystocia for the same gestation-length period (Supplemental Table S2). Once birth weight was included in the statistical model, no association existed between gestation length EBV and calving dystocia ($P = 0.33$).

DISCUSSION

Gestation length is particularly important for seasonal-calving herds in which cows are bred to calve at specific seasons to align with optimal grass growth for pasture-

based systems; each delayed day in calving date from the optimal date is thought to incur an average cost of €3.86 per cow in Ireland (Shalloo et al., 2014). Knowing the association between DIM and gestation length, along with other associations reported here and elsewhere, should help improve the prediction of expected calving date for management purposes. This can be simply achieved at the time of pregnancy diagnosis by summing the relevant model fixed effects solutions, including the most up-to-date estimate of genetic merit for the recorded sire; if, during pregnancy diagnosis, the sex of the calf is determined, as well as whether it is a singleton or twin, these effects can also be considered. Although numerous studies have explored the association between gestation length and various nongenetic factors in cattle (Norman et al., 2009; Tomasek et al., 2017; Vieira-Neto et al., 2017), as well as its associations with calf perinatal mortality and calving difficulty in dairy cattle (Atashi and Asaadi, 2019), substantial research gaps still exist. Notably, no study in cattle has investigated the association between gestation length and nongenetic factors while also accounting for estimated genetic merit for direct gestation length as a covariate in the statistical model. Furthermore, the association between gestation length and DIM at conception remains largely unexplored. In addition, few studies have attempted to associate gestation length with perinatal mortality, calving assistance and dystocia while also accounting for differences in calf birth weight. Exploration of associations conditional on

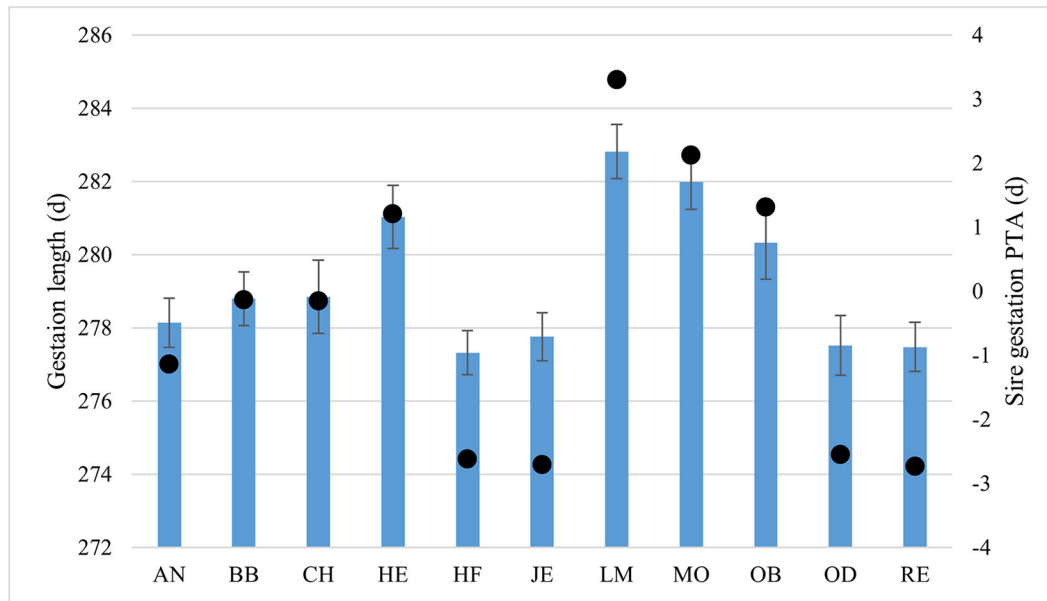


Figure 3. Predicted marginal means for gestation length (with error bar extending 1 SE on either side of the mean), represented as bars, along with mean PTA represented as black circles for different sire breeds (AN = Aberdeen Angus; BB = Belgian Blue; CH = Charolais; HE = Hereford; HF = Holstein-Friesian; JE = Jersey; LM = Limousin; MO = Montbéliarde; OB = other beef breeds; OD = other dam breeds; RE = Scandinavian Red).

differences in calf birth weight or genetic merit for direct gestation length (including sire breed) can provide additional information on the underlying association factors. Although genetic correlations between gestation length and calving performance have been documented in dairy cattle (Hansen et al., 2004; Nogalski and Piwczyński, 2012; Eaglen et al., 2013), genetic correlations describe the strength of linear associations between the variables of interest. Using calf estimates of genetic merit for gestation length, the present study explored whether nonlinear relationships also existed. In all, several factors were deemed to be associated with variability in gestation length in dairy cows, which, in turn, was associated with differences in calving performance. Calf birth weight explained some, but not all, of the associations between gestation length and calving assistance, calving dystocia, and perinatal mortality.

The mean gestation length in the sample population used in the present study of 280.6 d is comparable to, albeit slightly longer than, previously documented gestation lengths in dairy cow populations. Norman et al. (2009) reported a mean gestation length of 279.4 d based on over 11 million parturitions from 7 US dairy breeds. In a separate study, Johanson and Berger (2003) observed a mean gestation length of 277.9 d from 4,528 calving records of American Holstein dairy cows. Similarly, Hansen et al. (2004) reported a mean gestation length of 278.5 d from 184,000 Danish Holstein cows calving. In the present study, the perinatal mortality rate was 4.03%, which is lower than the 8.2% (Lombard et al.,

2007), 5.7% (Meyer et al., 2001), and 8% (Silva del Río et al., 2007) reported for US commercial dairy farms. Similarly, the dystocia rate of 4.42% observed in the present study is notably lower than 10.6% on a 4-point scale (Atashi et al., 2021) and 11.2% on a 5-point scale (Hansen et al., 2004) documented for Iranian and Danish Holstein-Friesian cattle, respectively. Nonetheless, the incidences of perinatal mortality and dystocia observed from the research farms in the present study are similar to those reported nationally in Ireland for perinatal mortality (4.3%; Mee et al., 2008) and dystocia (6.8%; Mee et al., 2011). Differences in incidence rates between populations are, nonetheless, not unexpected given the inter-herd and inter-population differences that are likely to exist in calving management, nutrition, genetics, and environmental factors. The incidence of twins reported in present study of 2.61% is lower than US values of 4.7% and 3.0% reported by Norman et al. (2009) for Brown Swiss and Holstein, respectively; this could be due, in part, to the lower milk yield of Irish cows, given its association with lower incidence of multiple ovulations (Carthy et al., 2016).

Days in Milk at Conception

No study has previously explored the association between DIM at conception and gestation length in cattle. The present study concluded that mean gestation length reduced with a delay in days after calving when conception occurred. For example, cows that conceived earlier

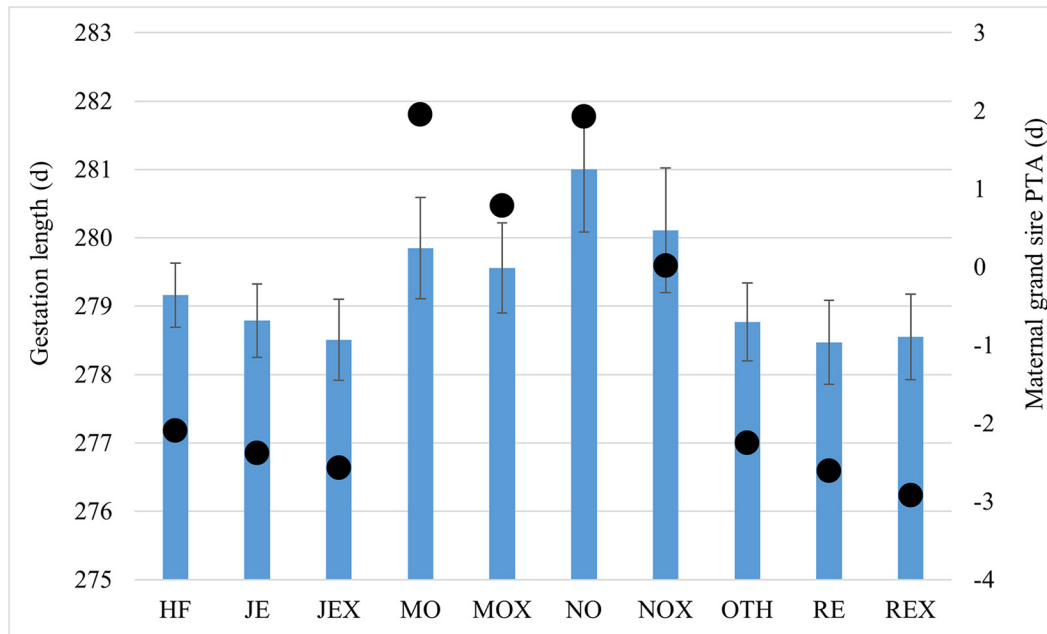


Figure 4. Predicted marginal means for gestation length (with error bar extending 1 SE on either side of the mean) for different dam breeds, represented as bars, and the maternal grandsire gestation PTA represented as black circles (HF = Holstein-Friesian; JE = Jersey; JEX = Holstein-Friesian × Jersey cross; MO = Montbéliarde; MOX = Holstein-Friesian × Montbéliarde cross; NO = Normande; NOX = Holstein-Friesian × Normande cross; OTH = other breeds; RE = Scandinavian Red; REX = Holstein-Friesian × Scandinavian Red cross).

(<49 DIM) had an average gestation length 1.5 d longer than those that conceived after 132 DIM. A similar phenomenon has been observed in horses, where mares mated at the end of the breeding season had a shorter gestation than those mated at the beginning of the season (Satué et al., 2011). One could speculate that the use of sires with shorter gestation length as the breeding season progresses may have contributed to the shorter gestation lengths observed in cows that conceive late in lactation; however, the trend with DIM in the present study persisted even when calf (or sire) genetic merit for gestation length was included in the model, suggesting that this is certainly not the only effect. Furthermore, artificial induction of parturition is not practiced in the research herds used in the present study and, therefore, is not a contributing factor to the shorter gestation lengths observed in cows conceiving after 132 DIM.

Despite the observed shortening in gestation length as DIM at conception increased, a positive association existed between DIM at conception and calf birth weight. In fact, calves born from conception <49 DIM were, on average, 0.8 kg lighter (i.e., 2% of mean birth weight) than those born following conception >132 DIM. Birth weight is positively genetically correlated with gestation length in cattle, with longer gestation periods associated with heavier birth weight (Coleman et al., 2021). Although beef bulls tend to be used at the end of the breeding season in Irish dairy herds (Berry

et al., 2014), and calves from beef bulls are, on average, heavier than those from dairy bulls (Condon et al., 2024), sire breed was included in the statistical model in the present study; therefore breed is not a contributing factor to this observation.

Parity, Twinning, and Calf Sex

Calvings from first-parity heifers were not included in the present study, because no reliable date of service was available; many of the matings to heifers were to natural-mating bulls. Nonetheless, the genetic correlation between gestation length in heifers and cows has already been documented to be close to unity in dairy cattle (0.88 to 0.95, Haile-Mariam and Pryce, 2019; 0.96 to 0.98, Norman et al., 2009). A difference of 1.3 d in mean gestation length existed between the parity with the shortest gestation length (i.e., parity 2) and that with the longest gestation length (i.e., parities 8 to 12). Corroborating the results from the present study, others have also documented longer gestations in older-parity cows (Norman et al., 2009; Nogalski and Piwczyński, 2012; Atashi and Asaadi, 2019) and camels (Mehta and Sahani, 2009; Nagy and Juhász, 2019). Although an association between parity and gestation also has been found in horses, the association was opposite, in that gestation was longer in younger mares (Satué et al., 2011). The observed increase in gestation

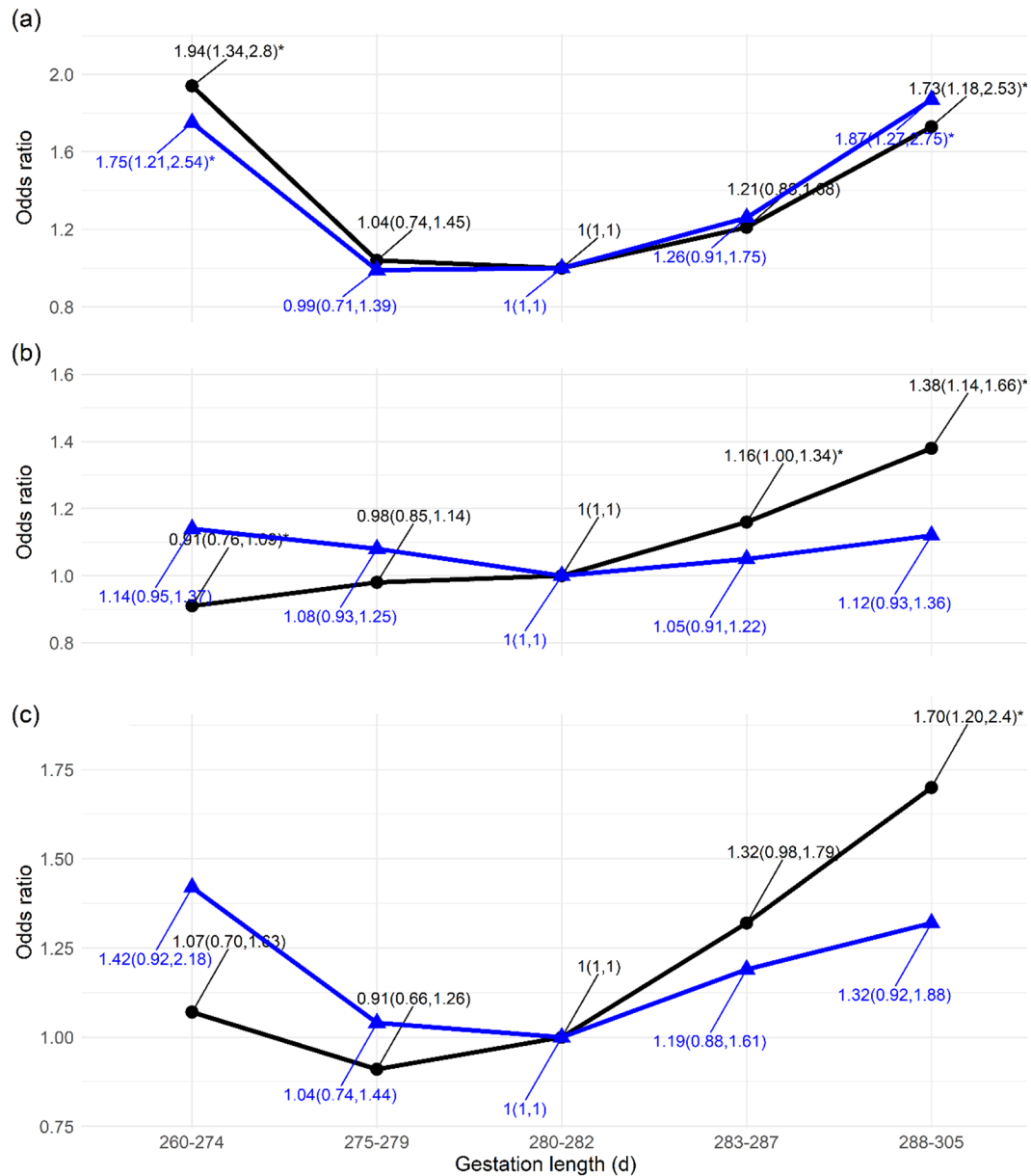


Figure 5. Odds ratios (95% CI in parentheses) for the associations between different categories of phenotypic gestation length and (a) perinatal mortality, (b) calving assistance, and (c) dystocia, relative to the reference gestation length of 280 to 282 d with (blue triangles) or without (black circles) adjustment for birth weight. *Odds ratio different ($P < 0.05$) to gestation period of 280 to 282 d.

length with parity in the present study is likely due to physiological maturity of the cow's reproductive system, whereby their bodies undergo adaptations that better support fetal development and longer gestation periods (Hansen et al., 2004).

Shorter gestations, by 3.9 to 5.9 d, for twins relative to singletons have already been reported in US dairy cows (Norman et al., 2009; Olson et al., 2009; Dhakal et al., 2013), consistent with the 4.3-d shorter gestation length estimated in the present study. The shorter gesta-

tion length in twin pregnancies is likely due to the larger combined size and weight of twin calves compared with single calves, which, in turn, exerts more pressure stimuli on the uterine wall, leading to earlier parturition (Olson et al., 2009). As expected, this shorter gestation of twins was also associated with a lighter birth weight of the individual calves, being, on average, 15% lighter than singletons, which is slightly lower than the 21% reported from 104,572 calving events in Iranian Holsteins (Hosseini-Zadeh, 2010). Shorter gestations and lighter

individual birth weights for multiple births relative to singletons have also been reported in other species such as humans (ESHRE Capri Workshop Group, 2010) and sheep (Hancock et al., 2012; McHugh et al., 2017).

A 1.0- to 1.4-d longer gestation length of male calves than heifer calves, consistent with the 1.25 d in the present study, has been reported in other international dairy cow populations (Hansen et al., 2004; Norman et al., 2009; Tomasek et al., 2017; Haile-Mariam and Pryce, 2019) and may be due to the longer development time required for male fetuses, influenced by their greater weight and growth demands, necessitating additional time in utero for optimal development.

Inter- and Intra-breed Associations

The longer mean gestation of beef sire breeds relative to dairy sire breeds has been documented elsewhere (King et al., 1985; Norman et al., 2009; Fitzgerald et al., 2015), although within-breed variability in genetic merit for gestation length per sire has also been clearly reported (Berry and Ring, 2020). Notably, the predicted marginal mean gestation length in the present study varied by sire breed, with the shortest gestation for Holstein-Friesian (277.3 d) and Jersey (277.8 d) and the longest for Limousin (282.8 d) and Montbéliarde (282.0 d). Additionally, the difference between the highest and lowest gestation length sire PTA across the entire data set was 12.83 d (SD = 1.94 d). This demonstrates the capacity of astute sire selection, especially toward the end of the breeding season, to help deliver a compact calving season.

Although animal-level factors associated with gestation length have been well publicized (Norman et al., 2009; Tomasek et al., 2017; Vieira-Neto et al., 2017), less well explored is the association between EBV for gestation length and phenotypic gestation length, as well as whether that association differs by animal-level factors. Results from the present study indicate that, within breed, phenotypic gestation length increased, on average, by 1.11 d per 1-d increase in calf EBV for gestation length, which is close to the expectation of 1. In fact, the regression coefficient was 1.06 d when neither sire nor dam breed was included in the model. Therefore, producers can be confident that across-breed genetic evaluations for gestation length, in Ireland at least, do capture the intra- and interbreed differences, although there does appear to be some residual interbreed sire effects over and above those accounted for by the across-breed genetic evaluation. Being able to fairly compare breeds on the same scale is important for bull selection, both when choosing among sires of different breeds and when comparing the PTA of the sires to that of the herd of females.

Gestation Length Associations with Calving Performance

The unfavorable influence on perinatal mortality in calves born from very short and very long gestation length has been previously documented in cattle (Hansen et al., 2004; Bleul, 2011; Jourdain et al., 2024), consistent with the trends observed in the present study. It is hypothesized that shorter gestation lengths may lead to smaller calves that are less likely to survive, whereas longer gestations can result in larger calves that face delivery complications and increased mortality rates (Stachowicz et al., 2023). However, adjusting the association for differences in birth weight in the present study had minimal effect on the odds ratios; the same was true when adjusting for direct calf EBV for gestation length. This therefore indicates that factors other than calf birth weight and genetic merit for gestation length influence the association between gestation length and perinatal mortality.

The association between gestation length and the need for calving assistance indicates that longer gestation periods (288 to 305 d) in particular are associated with a greater likelihood of calving assistance or dystocia. Specifically, a gestation length of 288 to 305 d was associated with 1.70 times the odds of experiencing dystocia relative to calves with a gestation length of 280 to 282 d. Several other studies, including those on cattle, have also reported a greater risk of calving complications following longer gestation length (Olson et al., 2009; Atashi and Asaadi, 2019; Basiel et al., 2024). Although producers frequently select sires based on breeding indexes favoring shorter gestation periods to reduce calving interventions (Berry et al., 2024), this strategy only partially mitigates calving difficulties, as shown in the present study. Nonetheless, incorporating both gestation length genetic merit and birth weight into the statistical model reduced the odds of calving assistance or dystocia for extreme gestation lengths (i.e., very short or very long), although they still had greater odds compared with average gestation length. Therefore, incorporating genetic merit for gestation length into breeding strategies, together with other factors such as birth weight, has the potential to minimize the need for calving assistance.

CONCLUSIONS

Several factors, namely parity, twin births, calf sex, month and year of birth, DIM at conception, both dam and sire breed, and within-breed genetic merit for direct gestation length were all associated with gestation length in dairy cows. Longer gestation length was, on average, associated with older cows, cows giving birth to singleton male calves, and cows conceiving early in

the lactation cycle. The study provides valuable insights for predicting expected calving dates by considering key factors associated with gestation length; the model solutions can easily be incorporated into decision support tools that predict herd calving patterns. Knowledge of these factors can help producers optimize herd average gestation length within the context of an overall breeding strategy, avoiding the potential pitfalls of extremely short gestation and the repercussions of same.

NOTES

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Nonstandard abbreviations used: AN = Aberdeen Angus; AI = artificial insemination; BB = Belgian Blue; CH = Charolais; HE = Hereford; HF = Holstein-Friesian; JE = Jersey; JEX = Holstein-Friesian × Jersey cross; LM = Limousin; MO = Montbéliarde; MOX = Holstein-Friesian × Montbéliarde cross; NO = Normande; NOX = Holstein-Friesian × Normande cross; OB = other beef breeds; OD = other dam breeds; OTH = other breeds; RE = Scandinavian Red; REX = Holstein-Friesian × Scandinavian Red cross.

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